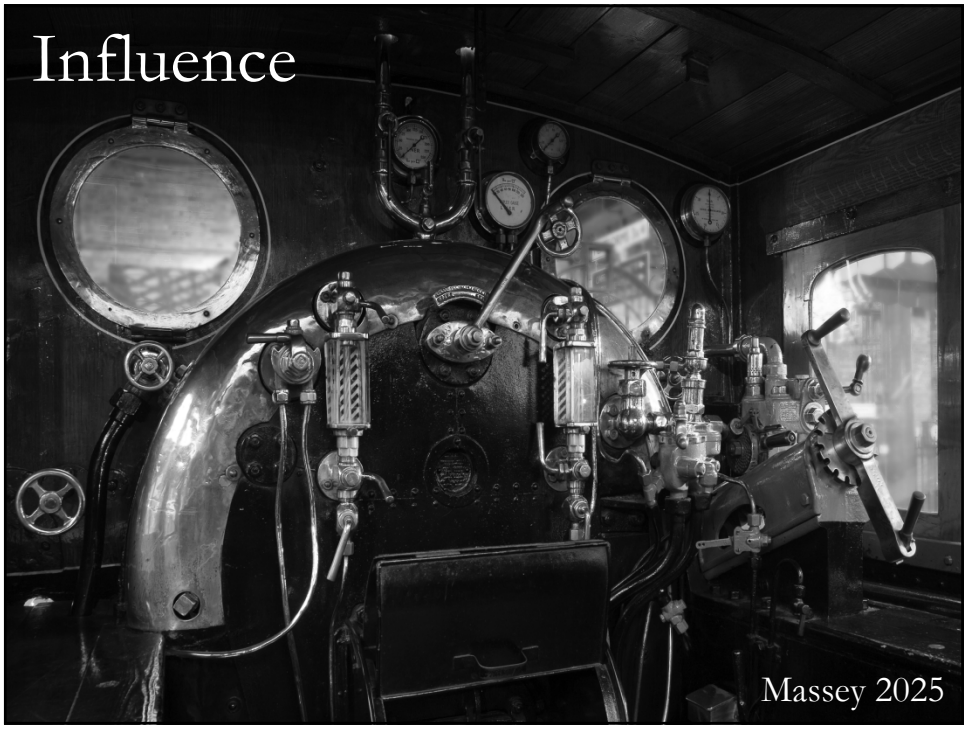


1

Today's Agenda

Application #1

Course Admin

Networks I

Intentionality

2

Course Admin

Next Class (Networks II)

- Heidi Roizen case
- Two readings – Susan Cain and David Schuppan
- Survey – due 8am on class day

Persuasive Video

- Next stage: Evaluate 7 videos
  - Email invitation with link coming tomorrow/Tuesday

Leverage Inventory

- Complete self assessment by Thursday, 11/13
- Make sure you have 8 evaluators in by Thursday, 11/13
  - You can continue to add/invite until then

3

Tracker: Gala Dinner seat sign-up form

List trackers

This form allows you to select a seat for Friday night's gala dinner. To use this form click on the table (the first column) you would like to be seated at, enter your name in the first available seat, and then press the Save button.

Choose a filter | filter

64 items found

a . b . c . d . e . f . g . h . i . j . k . l . m . n . o . p . q . r . s . t . u . v . w . x . y . z . All on Table

Table	Seat 1	Seat 2	Seat 3	Seat 4	Seat 5	Seat 6	Seat 7	Seat 8	Seat 9	
1	Chuck Weinberg	Joanne Weinberg	Dan Muzyka	Don Morrison	Sherie Morrison	Dave Montgomery	Toby Montgomery	Josh Eliashberg	Berend Wierenga	Tjitske Wierenga
2	Dan Putler	Rick Staelin	John Little	Arvind Rangaswamy	Rangaswamy guest	Rangaswamy guest	Tulin Erdem	Ashwin Malshe	Bryan Lilly	Berna Devezer
3	Fred Feinberg	Debu Purohit	Werner Reinartz	Franziska Völckner	Karen Gedenk	Rich Gonzalez	Elea Feit	Eric Bradlow	Judy Pastor	
4	John Hauser	Marija Hauser	Gui Liberali	Daria Silinskaia	Glen Urban	Theos Evgeniou	Lynd Bacon	Terry Elrod	Seenu Srinivasan	Kartik Hosanagar
5	Gary Lilien	Dorothy Lilien	Pam Morrison	John Roberts	Peter Danaher	P.V. (Sundar) Balakrishnan	Sengun Yeniyurt			
6	Sandy Jap	Qiong Wang	Scott Fay	Federico Rossi	Bruce Clark	Prasad Naik	Ernan Haruvy	Xin Wang	Ross Rizley	Shuba Srinivasan
7	Vithala Rao	Meng Su	Anand Bodapati	Venky Shankar	Vishal Narayan	Girish Mallapragada	Sungho Park	Sanjeev Tripathi	Anirban Mukherjee	Richard Batsell
8	Oliver Heil	Shyda Valizade	Mark Elsner	Sven Theysohn						
9	Destan Kandemir	Burcu Tasoluk	Wenbin Sun							
10	Raj Grewal	Min Ding								
11	Peter Pal Zubcsek	Zsolt Katona	Dina Mayzlin	Luc Wathieu	Catherine Tucker	Anja Lambrecht	Robert Zeithammer	Koen Pauwels	Juanjuan Zhang	Avi Goldfarb
12	Chakravarthi Narasimhan	Sanjay Dhar	Pavan Chennamaneni	Rajesh Chandy	Thorsten Wiesel	Sonja Gensler	Corine Noordhoff	Clemens Köhler	Nikos Kalogeras	
13	Miguel Villas-Boas	Sofia Villas-Boas	Dmitri Kuksov	Ying Xie	Liang Guo	Thanh Tran				
14	Guy Pearce									
15	Santiago Caballero	Hui-I Yao								
16	Horst Treiblmaier	Annabelle Griego								
17	Doug MacLachlan	Natelle MacLachlan								
18	Jenny van Doorn	Susanne Stingel	Robert Wilken	Sara Kremer	Tjitske Rispens	Ernst Osinga	Erjen van Nierop	Janny Hoekstra	Tammo Bijmolt	Hans Risselada
19	Jaap Wieringa	Wander Jager	Peter Verhoef	Frank van Rijnsoever	Pascal van Hattum	Marieke Ruijzing	Subom Rhee			
20	Hsiu-Wen Liu	Yu-Li Lin	Ting-Ling Lin	Ching-I Chen	Kung-Hsin Shao	Chih-Ping Wang				
21	Sanjog Misra	Wesley Hartmann	Paul Nelson	Paul Neson (guest)	Bart Bronnenberg	Paul Ellickson	Günter Hitsch	Dan Horsky	Sharon Horsky	Paulo Albuquerque
22	oliver rutz	Mitch Lovett	Song Yao	Woochoel Shin	Catarina Sismeiro	Michael Trusov	Jorge Silva-Risso	Adriana Gigliotti	Carl Mela	Amit Joshi

4

Connects to some general questions

Should I go to the gala?

- I.e., How much should I network?

Who should I sit with?

- I.e., what should my network look like?

What am I going to talk about?!

- I.e., how do I build relationships?

5

Networks Topic Outline

Monday: Networks I

- Social Capital introduction
- Weak ties
- Structural holes – theory
- Structural holes – practice

Wednesday: Networks II

- Heidi Roizen case
- Practitioner tips
- Reciprocity
- Wrap-up

6

Networks Topic Outline

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- Wrap-up

Network "use cases"

Personal: job opportunities, restaurant recommendations, good tickets, etc.

Professional

- Finding qualified candidates for a job
- New deal opportunities
- Understanding why and how a decision was made
- Learning about a new technology
- Getting an expert opinion

Networks as Social Capital

Social capital parallels Human capital

- You build it over time
- It is part of what you contribute to a team
- It is a reason firms hire you

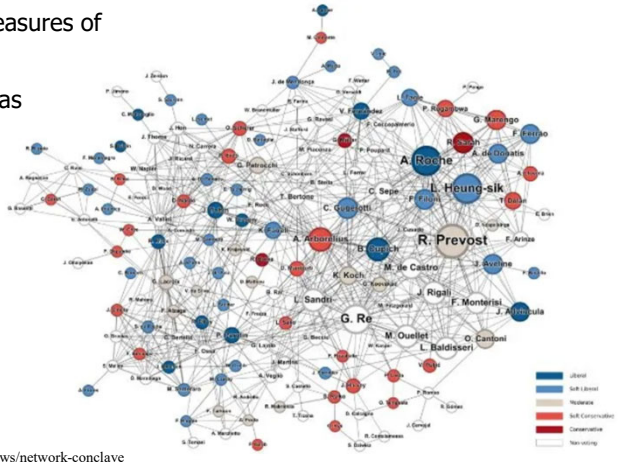
9

Pope Leo's network was an advantage

Bocconi study before the conclave

3 independent measures of network ties

Prevost ("Leo") was highest on status.



https://www.unibocconi.it/en/news/network-conclave

10

Short-term / Long-term Problem

A year from now, or 5 years from now, you'd rather have a good network than a weak network. You see that clearly and prefer it.

But today you're busy, you'd rather stay at your desk and get work done.

Trouble is, tomorrow you will feel the same way. And the next day, and so on, for another year (or 5).

What to do about this?

11

Short-term / Long-term Problem

Classic study:

What would you like with your sandwich [next Thursday / right now]?



vs.



These are "time-inconsistent preferences"

- Like savings and exercise (and healthy eating)

12

Short-term / Long-term Problem

These are "time-inconsistent preferences"

- If you let short-term preferences ("myopia") guide behavior every day you will make sub-optimal decisions for the long-term.
- Rx: A policy
  - A course of action chosen (intentionally, from a cool, rational distance) that guides behavior each day
  - **A possible heuristic policy: Invest more in your professional network than is convenient.**

Okay, so how?

Two big and very different questions:

1) Macro: Optimal network structure?
   Today's class

2) Micro: Effective relationship building?
   Next class

This week's themes

Your network is a vital part of how you perform at work

Your network is shaped by decisions you make, big and small, every day.

Bringing even just a little more thoughtfulness to your relationship-building decision making can make a huge difference.

Being thoughtful about your network does not mean losing your integrity or genuine-ness.

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Networks Topic Outline

Monday: Networks I

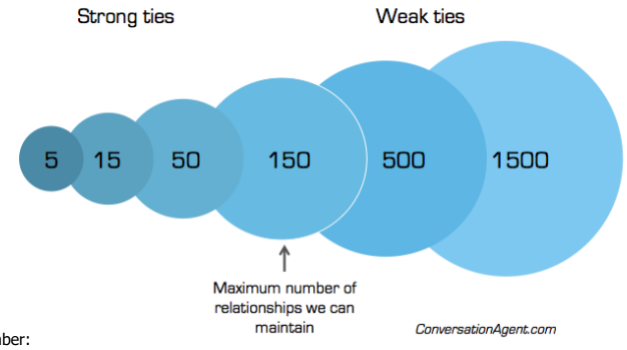
- Social Capital introduction
- **Weak ties**
- Structural holes – theory
- Structural holes – practice

Wednesday: Networks II

- Heidi Roizen case
- Practitioner tips
- Reciprocity
- Wrap-up

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Strong ties vs. Weak ties



Dunbar Number:
"this limit is a direct function of relative neocortex size, and that this in turn limits group size … the limit imposed by neocortical processing capacity is simply on the number of individuals with whom a stable inter-personal relationship can be maintained."

17

Granovetter 1973

The Strength of Weak Ties[1]

Mark S. Granovetter
Johns Hopkins University

Analysis of social networks is suggested as a tool for linking micro and macro levels of sociological theory. The procedure is illustrated by elaboration of the macro implications of one aspect of small-scale interaction: the strength of dyadic ties. It is argued that the degree of overlap of two individuals' friendship networks varies directly with the strength of their tie to one another. The impact of this principle on diffusion of influence and information, mobility opportunity, and community organization is explored. Stress is laid on the cohesive power of weak ties. Most network models deal, implicitly, with strong ties, thus confining their applicability to small, well-defined groups. Emphasis on weak ties lends itself to discussion of relations *between* groups and to analysis of segments of social structure not easily defined in terms of primary groups.

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Strong ties vs. Weak ties

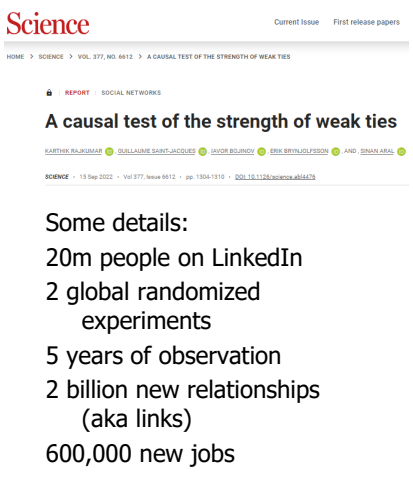
What makes a strong tie?

The-more-the-better of:
Time
Intensity
Trust

What makes a weak tie?

Occasional interactions
Prompted by context
Facilitated by existing connections
Much lower maintenance

And a surprisingly effective source of information & opportunities


Science

Current Issue First release papers

HOME > SCIENCE > VOL. 377, NO. 6612 > A CAUSAL TEST OF THE STRENGTH OF WEAK TIES

REPORT SOCIAL NETWORKS

A causal test of the strength of weak ties

KARTHIK RAJKUMAR, GUILLAUME SAINT-JACQUES, IAVOR BOJINOV, ERIK BRYNJOLFSSON, AND SINAN ARAL

SCIENCE • 15 Sep 2022 • Vol 377, Issue 6612 • pp. 1304-1310 • DOI: 10.1126/science.abl4476

Some details:

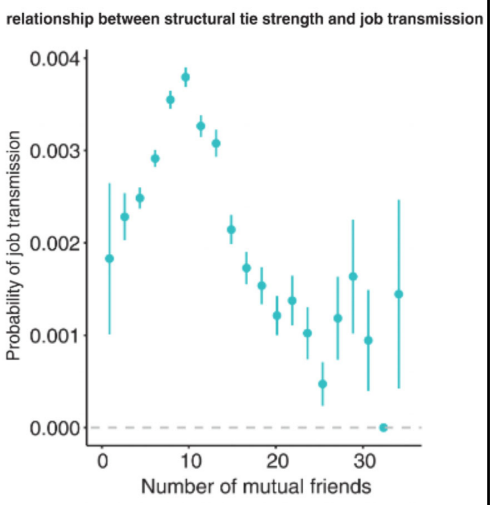
20m people on LinkedIn

2 global randomized experiments

5 years of observation

2 billion new relationships (aka links)

600,000 new jobs



Networks Topic Outline

Monday: Networks I

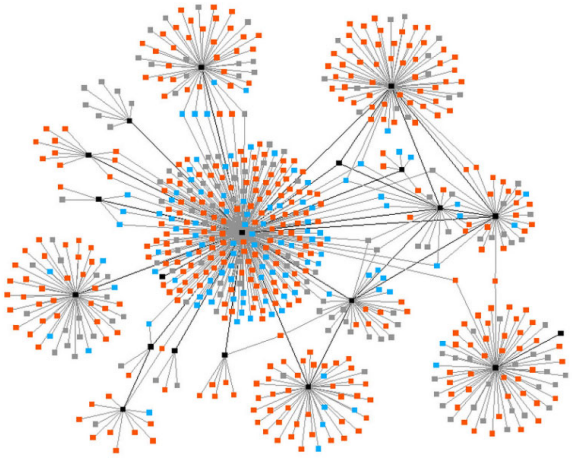
- Social Capital introduction
- Weak ties
- **Structural holes – theory**
- Structural holes – practice

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Optimal network? Optimal position?



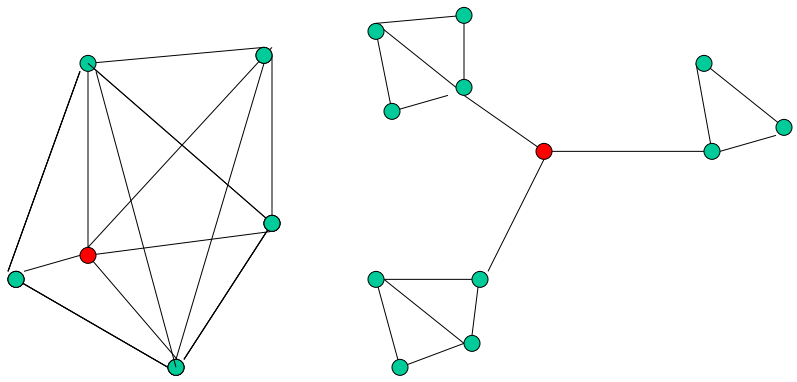
22

Optimal network position?

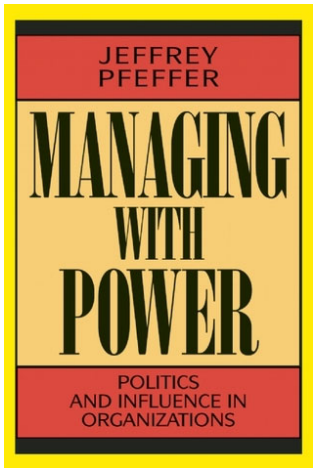
What's the best structure? Position?

- Central position?
  - Perceived as leader, indispensable
- Large number of ties?
  - More contacts for information, help
- Strong ally?
  - Advocate in hiring and promotion; provides training, information, access to resources
- Cohesive network?
  - Value of cohesion
- **Structural hole?**
  - **Centrally placed with ties to people who are <u>not</u> cohesive (i.e., <u>not</u> tied together)**

Which position would you rather hold?



Resource Dependence



"Power is vested in us by the dependence of others, and that dependence is a function of how much others need what we control, as well as how many alternative sources for that resource there are."

- Pfeffer -

25

Resource Dependence

Consider Panama…



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Resource Dependence



"In order to be irreplaceable, one must always be different."

- Coco Chanel

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"Model collapse" (Ian Leslie, 10/25/25)

A machine learning model gets trained on human data - photographs; Reddit forums; Amazon reviews; books and music and so on. We're already at the stage where the big models have hoovered so much of this data into their maws that finding fresh sources to train on is hard. AI models have no problem *producing* data, however, so the obvious answer is to train on data from other models. Indeed this is inevitable, as the internet becomes swamped by LLM-generated text and images.

But there's a problem with this. Model-generated data tends to be more predictable and less diverse than human data, as it imposes statistical patterns on the infinite variety of human outputs. Over time, this creates a feedback loop. Each new iteration of the model inherits the biases and errors of the previous one, but with less variety and signal. Eventually, the model converges towards - or collapses into - the generic and repetitive. It loses accuracy, nuance, creativity until only a thin and colourless monoculture remains. This is 'model collapse'.

Damn those machines, erasing the incorrigible plurality of human minds! And yet I think Karpathy is right when he points out this is also something we do to ourselves: **"Humans collapse during their course of their lives."** We, too, learn patterns in the data of experience on which we then become over-dependent, favouring the model in our heads over the flux of reality. As we get older, we become rigid in our thinking, set in our ways. We talk to the same friends and use the same information sources. We return to the same old thoughts and say the same old stuff.

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Can we make these ideas more precise?

29

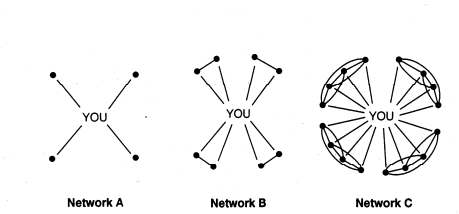
Modeling Networks

To model we must simplify, and understand the objectives and constraints.

- What are the benefits of a network?
- What are the costs?

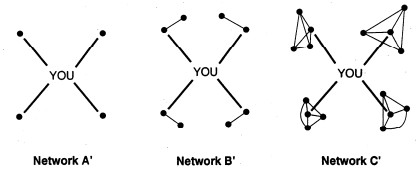
30

Redundant vs. Efficient Ties



Redundant Ties

Increasing Information Sources →

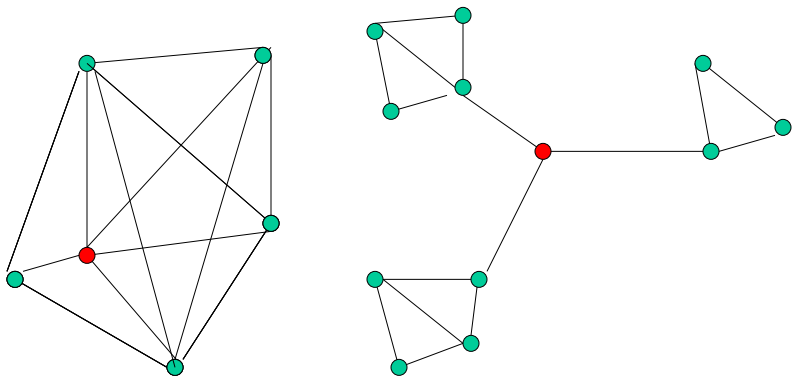


Efficient Ties

Assumption: Resources needed to maintain relationships (e.g., time, emotion) are scarce

31

Dense vs. Disconnected



What happens if you remove the red point in each network?

32

Features of Different Network Structures

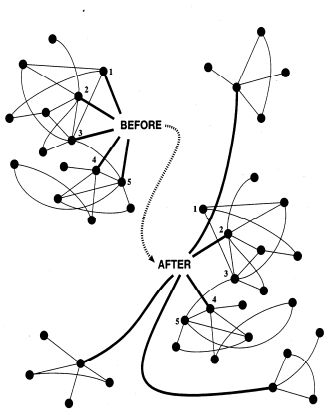
Dense Structure (Few Holes)

- People are highly interconnected with mutual influence
- Often reflects cohesive, very similar groups
- Often reflects groups who work closely together, think alike, and thus hold redundant information
- Often tap a restricted set of resources

Disconnected Structure (Many Holes)

- Often reflects non-cohesive, heterogeneous groups
- Provide opportunities in terms of differentiated, non-redundant information
- Provide opportunities for accessing a wide array of resources
- Research and theory by sociologist Ron Burt, building on weak-ties research by Mark Granovetter

Network Diversification



After diversification…

- Same number of direct ties
- Larger network (more secondary contacts)
- More diverse information sources

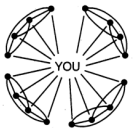
Benefits? Person filling hole…

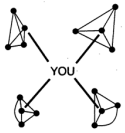
- Will be "first to know" information outside each pocket—broker information
- Can arrange partnerships
- Can play rival factions against each other
- Will be sought out for contacts

Some Limitations…

Shortcomings of being in a disconnected structure?

Benefits of being in a dense structure?





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E.g., network research using email

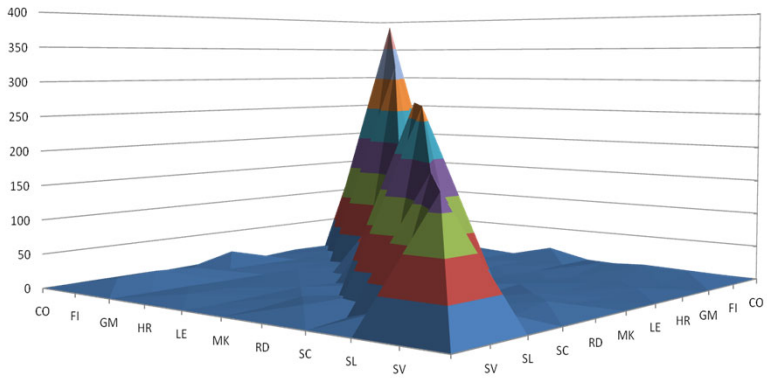
Adam Kleinbaum, Dartmouth (2010 job talk)

Large information technology company

- 30 product divisions, global presence
- Sampled all email activity for 30,328 employees over a 3-month period

37

Average E-mails Exchanged Across Job Functions

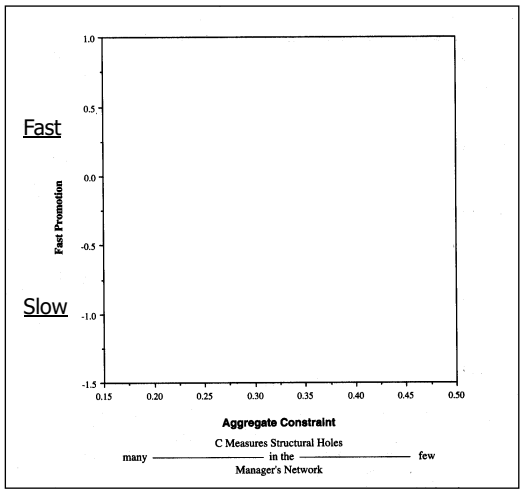


Virtually all email communication takes place within business unit!

38

"Returns" to Disconnected Structures

Rate of Promotion
(controlling for other factors, e.g., demographics, performance, etc.)



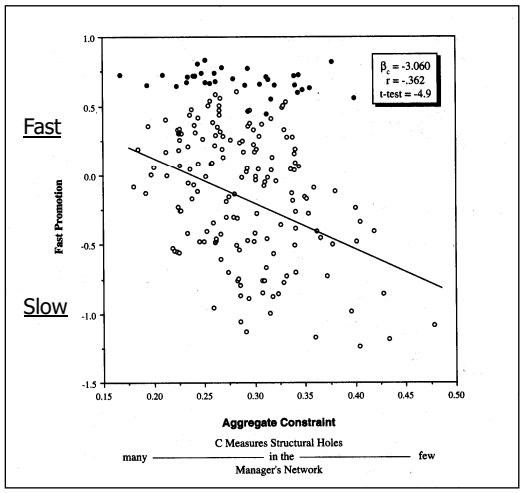
Research by Ron Burt

Many Disconnects Few Disconnects

39

"Returns" to Disconnected Structures

Rate of Promotion
(controlling for other factors, e.g., demographics, performance, etc.)



Research by Ron Burt

Many Disconnects Few Disconnects

40

The value of diverse inputs, extended

Parise, Whalen & Todd, "How Twitter Users Can Generate Better Ideas." MIT Sloan Management Review, Summer 2015.

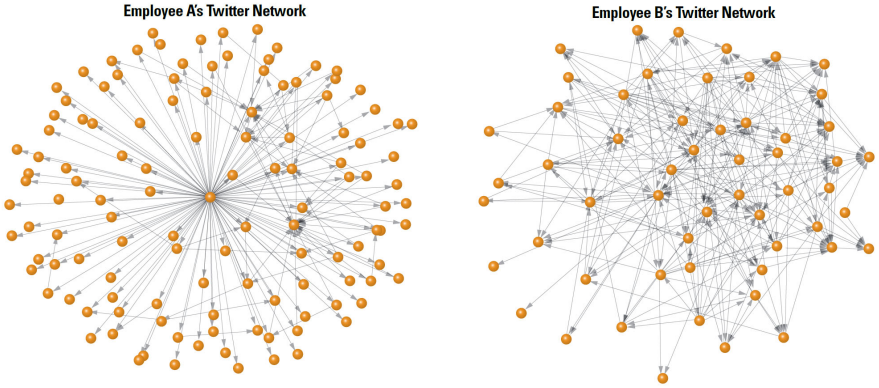
Organizational network analysis of 10 work groups in 5 organizations (consulting, tech, mfg, consumer food, higher ed)

Their interest: Employee innovation, both quantity and quality.

41

COMPARING TWO TWITTER NETWORKS

Although employees A and B follow approximately the same number of Twitter accounts, A's network is far more diverse than B's. For the most part, the people employee A follows are not following each other, which is more conducive to innovation and better idea generation. Compact Twitter networks like employee B's provide redundant information and are less conducive to ideation.



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Parise, et al's Conclusion

"Our research found that loose Twitter networks, such as employee A's, are better for ideation, because the potential for accessing a divergent set of ideas is greater."

Routine decisions => Network consequences

Would you want the following? Why or why not?

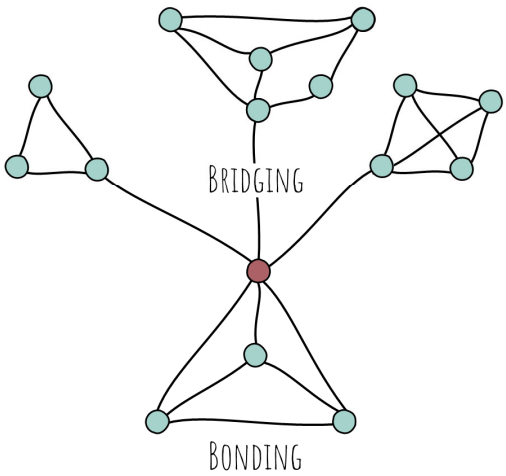
- A desk by the bathroom?
- An overseas assignment?
- A seat on a task force or committee?

Breakout Group Discussion

How can **you** diversify your professional relationships, and/or the information you're exposed to? Be specific/concrete.

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Strong ties and Weak ties matter



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Today's Agenda

Course Admin

Application #1

Gala Dinner exercise
Networks I

Intentionality

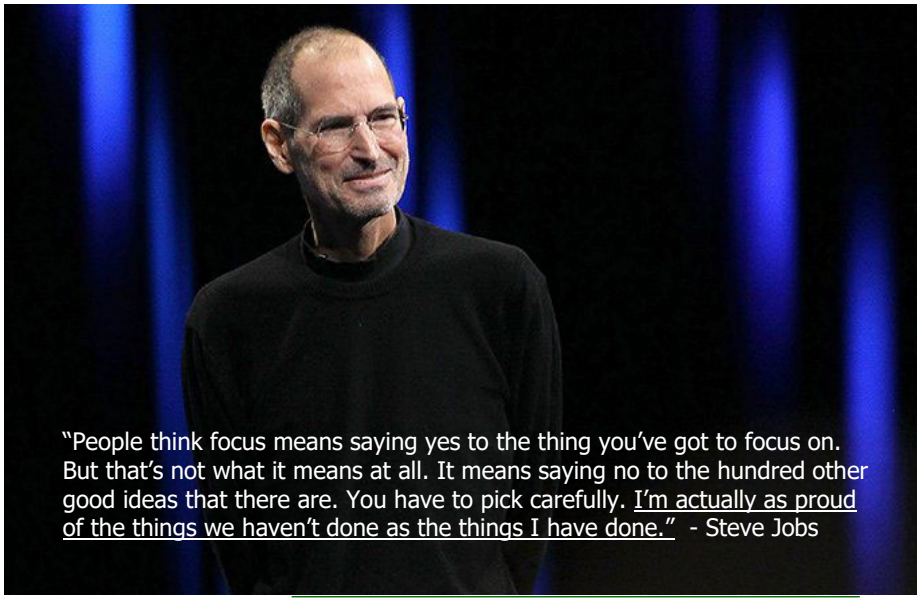
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On Intentionality

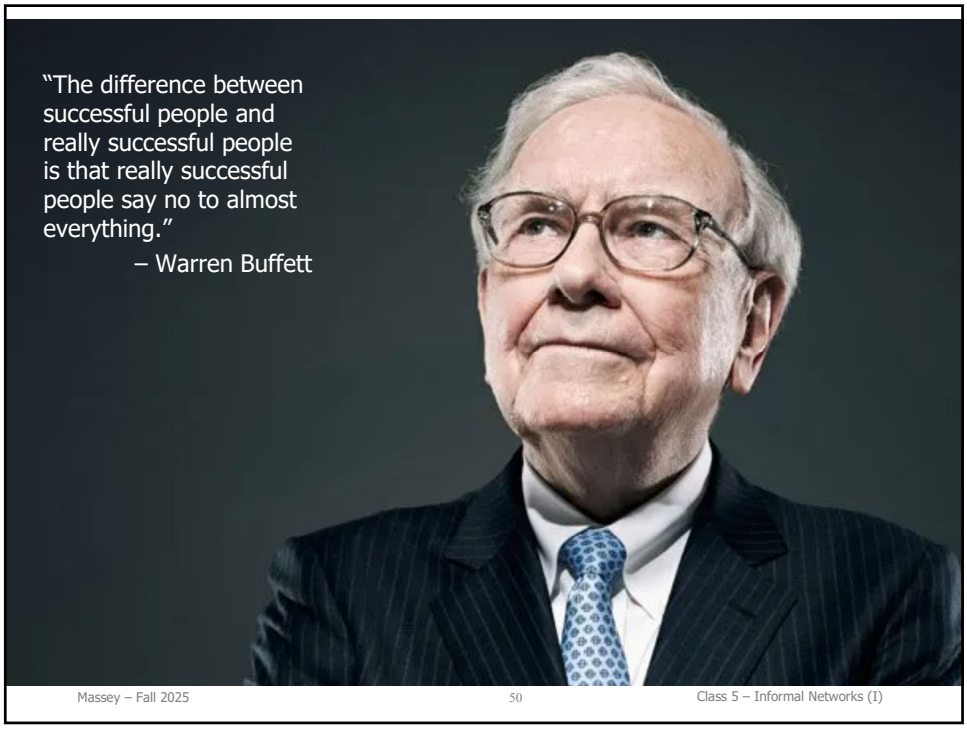
Definition: Acting with a goal in mind

Heuristic: You're not truly committed to a goal unless you're sacrificing lesser goals.

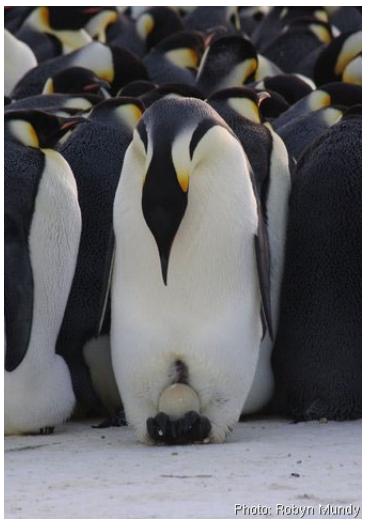
48


"People think focus means saying yes to the thing you've got to focus on. But that's not what it means at all. It means saying no to the hundred other good ideas that there are. You have to pick carefully. I'm actually as proud of the things we haven't done as the things I have done." - Steve Jobs

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"The difference between successful people and really successful people is that really successful people say no to almost everything."
– Warren Buffett

50



Those who reach greatness on earth reach it through concentration.

- Upanishads

51



52

Steinbeck, on his expedition to the Sea of Cortez, 1940:

"Hitler marched into Denmark and into Norway, France had fallen, the Maginot Line was lost – we didn't know it, but we knew the daily catch of every boat within four hundred miles. It was simply a directional thing; a man has only so much."

The Log from the Sea of Cortez (Ch 1), 1941/1951



Warren Buffet's "20 slot" rule

Via Charlie Munger, Stanford Business School, 1994:

"When Warren lectures at business schools, he says, 'I could improve your ultimate financial welfare by giving you a ticket with only 20 slots in it so that you had 20 punches—representing all the investments that you got to make in a lifetime. And once you'd punched through the card, you couldn't make any more investments at all.'

"He says, 'Under those rules, you'd really think carefully about what you did and you'd be forced to load up on what you'd really thought about. So you'd do so much better.'

"Again, this is a concept that seems perfectly obvious to me. And to Warren it seems perfectly obvious. But this is one of the very few business classes in the U.S. where anybody will be saying so. It just isn't the conventional wisdom.

"To me, it's obvious that the winner has to bet very selectively. It's been obvious to me since very early in life. I don't know why it's not obvious to very many other people."

E.g., VCs who lose focus suffer

Journal of Business Venturing Insights 16 (2021) e00266

Contents lists available at ScienceDirect

Journal of Business Venturing Insights

journal homepage: www.elsevier.com/locate/jbvi

The effect of venture capitalists straying from their industry comfort zones

Tyler J. Hull, PhD

College of Management, University of Massachusetts Boston, MA, 02125, USA

ARTICLE INFO

Keywords:
Venture capital
Specialization
Industry experience

ABSTRACT

I empirically test and find evidence that venture capital industry experience is more informative and impactful in venture capital investment when measured at the low non-aggregated industry level. Additional evidence shows that venture capital investments made outside of a venture capitalist's preferred investment industry suffer from significant decreases in exit success likelihood. This effect becomes stronger the more distant the investment industry is from the venture capitalist's preferred investment industry.

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Appendix

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On Intentionality

Soren Kierkegaard:

A saint is "someone who can will the one thing".

(See the appendix below for a meditation on this quote by Ronald Rolheiser, in which he contrasts Mother Teresa with Janis Joplin!)

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Rolheiser, from Holy Longing

"A saint is someone who can, precisely, channel powerful eros in a creative, life-giving way. **Søren Kierkegaard once defined a saint as someone who can will the one thing.** Nobody disputes that Mother Teresa did just that, willed the one thing—God and the poor. She had a powerful energy, but it was a very disciplined one. Her fiery eros was poured out for God and the poor. That—total dedication of everything to God and poor—was her signature, her spirituality. It made her what she was. Looking at Janis Joplin, the rock star who died from an overdose of life at age twenty-seven, few would consider her a very spiritual woman. Yet she was one. People think of her as the opposite of Mother Teresa, erotic, but not spiritual. Yet Janis Joplin was not so different from Mother Teresa, at least not in raw makeup and character. She was also an exceptional woman, a person of fiery eros, a great lover, a person with a rare energy. Unlike Mother Teresa, however, Janis Joplin could not will the one thing. She willed many things. Her great energy went out in all directions and eventually created an excess and a tiredness that led to an early death. But those activities—a total giving over to creativity, performance, drugs, booze, sex, coupled with the neglect of normal rest—were her spirituality. This was her signature. It was how she channeled her eros. In her case, as is tragically often the case in gifted artists, the end result, at least in this life, was not a healthy integration but a dissipation. She, at a point, simply lost the things that normally glue a human person together and broke apart under too much pressure. Looking at Joplin's life, and at our own lives, there is an interesting reflection to be made on Kierkegaard's definition of being a saint—someone who can will the one thing. Most of us are quite like Mother Teresa in that we want to will God and the poor. We do will them. The problem is we will everything else as well. Thus, we want to be a saint, but we also want to feel every sensation experienced by sinners; we want to be innocent and pure, but we also want to be experienced and taste all of life; we want to serve the poor and have a simple lifestyle, but we also want all the comforts of the rich; we want to have the depth afforded by solitude, but we also do not want to miss anything; we want to pray, but we also want to watch television, read, talk to friends, and go out. Small wonder life is often a trying enterprise and we are often tired and pathologically overextended**. Medieval philosophy had a dictum that said: Every choice is a renunciation.** Indeed. Every choice is a thousand renunciations. To choose one thing is to turn one's back on many others. To marry one person is to not marry all the others, to have a baby means to give up certain other things; and to pray may mean to miss watching television or visiting with friends. This makes choosing hard. No wonder we struggle so much with commitment. It is not that we do not want certain things, it is just that we know that if we choose them we close off so many other things. It is not easy to be a saint, to will the one thing, to have the discipline of a Mother Teresa. The danger is that we end up more like Janis Joplin; good-hearted, highly energized, driven to try to drink in all of life, but in danger of falling apart and dying from lack of rest."

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Networks, Creativity, and Time: Staying Creative through Brokerage and Network Rejuvenation

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NETWORKS, CREATIVITY, AND TIME:
STAYING CREATIVE THROUGH BROKERAGE AND NETWORK REJUVENATION

ABSTRACT

In this paper we adopt a dynamic perspective on networks and creativity to propose that the oft-theorized creative benefits of open networks and heterogeneous content are less likely to be accrued over time if the network is stable. Specifically, we hypothesize that open networks and content heterogeneity will have a more positive effect on creativity when network stability is low. We base our prediction on the fact that over time network stability begets cognitive rigidity and social rigidity, thus limiting individuals' ability to make use of the creative advantages provided by open networks and heterogeneous content. On the contrary, new ties bring a positive "shock" that pushes individuals in the network to change the way they organize and process knowledge, as well as the way they interact and collaborate – a shock that enables creators to accrue the creative advantages provided by open network structures and heterogeneous content. We test and find support for our theory in a study on the core artists who worked on the TV series *Doctor Who* between 1963 and 2014.

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Social Network Effects on Productivity and Job Security: Evidence from the Adoption of a Social Networking Tool

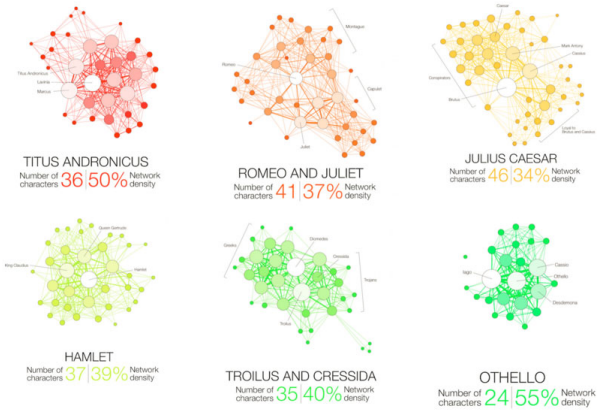
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By studying the change in employees' network positions before and after the introduction of a social networking tool, I find that information-rich networks (low in cohesion and rich in structural holes), enabled by social media, have a positive effect on various work outcomes. Contrary to the notion that network positions are difficult to alter, I show that social media can induce a change in network structure, one from which individuals can derive economic benefits. In addition, I consider two intermediate mechanisms by which an information-rich network is theorized to improve work performance—information diversity and social communication—and quantify their effects on productivity and job security. Analysis shows that productivity, as measured by billable revenue, is more associated with information diversity than with social communication. However, the opposite is true for job security. Social communication is more correlated with reduced layoff risks than with information diversity. This, in turn, suggests that information-rich networks enabled through the use of social media can drive both work performance and job security, but that there is a trade-off between engaging in social communication and gathering diverse information.

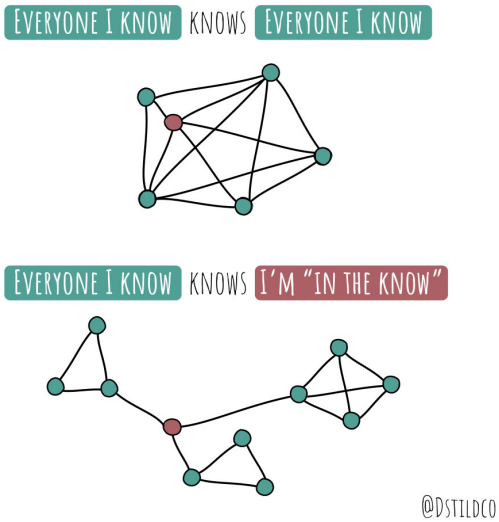
Key words: social media; social network; productivity; job security; information diversity; social communication; knowledge management
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E.g., the networks in Shakespeare



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